

The Single-Site T-Matrix of a Legendre Cell in Closed Form

Elastic Waves, Every Power of the Frequency on Six Constants

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Abstract

A volume-integral method for elastic waves divides a heterogeneous medium into cells, each with a single-site T -matrix that maps the field incident on it to the sources it radiates. The governing equation is written in displacement; how the strain inside a cell is represented is a modelling decision, and the original approximation takes it as uniform. We take the cell whose displacement and strain are independent Legendre expansions over the cube, tested against the same polynomials (a Galerkin cell), because of the representations in use it gives the most accurate coupling between cells: its interactions are exact double integrals over both cells, where an expansion about the cell's centre treats the receiving cell as a point. For the linear Legendre cell with a uniform contrast we give the single-site T -matrix in closed form at every power of the wavenumber. Its self-interaction is a power series, $\mathbf{K} = \sum_n (k_S h)^n \mathbf{K}_n$ with h the half-width, whose coefficients $\mathbf{K}_n$ are moments of powers of the distance over the cell and depend on neither the frequency, the cell size nor the medium. Every coefficient is exact: an even power of $k_S h$ is a combination with rational coefficients of six constants, 1 , π , $\sqrt{2}$, $\sqrt{3}$, $\ln(1 + \sqrt{2})$ and $\ln(2 + \sqrt{3})$, the last two the logarithms of the fundamental units of $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{3})$, and an odd power, which carries the radiation, is rational. The first of these is the radiation damping of a point force, recovered exactly. On the irreducible representations of the cube group the 36 unknowns of the cell separate into blocks of at most four, each inverted in closed form: the T -matrix is an explicit rational function of the contrast and of $k_S h$, whose poles are written down, and whose static dilatational block is Eshelby's result for a sphere. Through $(k_S h)^6$ the closed form reproduces the T -matrix computed by quadrature to 3.5×10^{-9} at $k_S h = 0.2$, falling as $(k_S h)^7$ as the next term predicts, and each of its 223 moments to 1.7×10^{-15} . Because the coefficients depend on neither the frequency, the cell size nor the medium, a frequency sweep integrates nothing at any frequency: the T -matrix costs 0.3 ms per frequency against 2 s by quadrature, after a one-off derivation of 30 s.

1 Introduction

A volume-integral method for elastic scattering replaces a heterogeneous medium by cells, here cubes, and couples them through the Green's tensor of the background. Each cell is characterised by its single-site T -matrix: the map from the field incident on it to the sources it radiates when

it is alone. The coupled system is then the multiple-scattering problem of the cells [1–4], and everything that a cell contributes to the solution, beyond its coupling to the others, is in its T -matrix.

How the field inside a cell is represented decides what that T -matrix is. The governing equation is written in displacement, and the strain enters it only through the stiffness contrast; how the strain is represented within a cell is therefore a modelling decision. The original approximation, of the equivalent-inclusion method [5, 6] and of the coupled-dipole method for elastic waves [7], takes it as uniform. That approximation can be relaxed in two ways. In the first, the displacement is expanded about the cell’s centre and the strain derived from it; the equation and its derivatives are imposed at the centre, which is natural for a single scatterer, whose T -matrix is then a property of the body alone [8]. In the second, the strain is an unknown in its own right, with its own polynomial expansion over the cell, and the equation is tested against the cell’s polynomials (a Galerkin cell). Which of the two a tiling of cells needs is measured on a layer, which has an exact solution (Table 2): with polynomials of degree p , an independent strain converges at order $2p + 2$, while a strain derived from each cell’s displacement converges at order 2, 2 and 4 for $p = 1, 2$ and 3.

This paper takes the second, because it gives the more accurate representation of the coupling between cells. Its interactions are exact double integrals over both cells, which are evaluated to round-off even for cells that touch [9]; an expansion about the centre treats the receiving cell as a point. On a cube cut into n^3 cells, where the field follows the corners, the first-moment Legendre cell of 36 unknowns is seven to twelve times more accurate than the third-gradient expansion about the centre with 60; on a smoothly graded sphere both converge at fourth order, and the Legendre cell is the more accurate for a given number of unknowns [8]. And a Galerkin cell holds the projection of the field and of the medium, which is what makes its error a projection error, known before the solve [10].

The single-site T -matrix of the linear Legendre cell was given in Nestor [9] in its static limit, entry by entry on five constants. Here it is given at every power of the frequency. The central result is that the self-interaction of the cell, the only part of its T -matrix that is an integral, is

$$\mathbf{K} = \sum_{n \geq 0} (k_S h)^n \mathbf{K}_n, \quad \mathbf{K}_n \text{ exact for every } n, \quad (1)$$

a power series in $k_S h$ whose coefficients are universal moments of powers of the distance over the cell [9], independent of the frequency, the cell size and the medium. We evaluate every one of them exactly (§3), and find that each even power of $k_S h$ is a combination with rational coefficients of the same six constants and that each odd power, which carries the radiation, is rational (§4). Computed once and stored, the $\mathbf{K}_n$ give the T -matrix at any frequency without an integral (§7). The cube group reduces the 36 unknowns to blocks of at most four (§5, Figure 1), on which the T -matrix is a few closed-form matrices per power. Section 6 gives the checks, each against an evaluation that shares no step with the closed form.

2 The cell and its single-site matrix

The governing equation. With time dependence $e^{-i\omega t}$, split the density and the stiffness into the background and the contrast, $\rho = \rho_0 + \Delta\rho$ and $c = c^0 + \delta c$, $\delta c_{bcde} = \Delta\lambda \delta_{bc}\delta_{de} + \Delta\mu(\delta_{bd}\delta_{ce} + \delta_{be}\delta_{cd})$. The contrast acts on the background as a body force, $\omega^2\Delta\rho u_b + \partial_c m_{bc}$ with $m_{bc} = \delta c_{bcde}\varepsilon_{de}$, and with $\mathbf{G}$ the background's Green's tensor the displacement is

$$u_a(\mathbf{x}) = u_a^0(\mathbf{x}) + \int G_{ab}(\mathbf{x} - \mathbf{y}) \omega^2 \Delta\rho u_b(\mathbf{y}) d^3y + \int \frac{\partial G_{ab}}{\partial x_c}(\mathbf{x} - \mathbf{y}) m_{bc}(\mathbf{y}) d^3y, \quad (2)$$

the moment radiating through the receiver derivative of $\mathbf{G}$ [8]. The strain enters only through m . Differentiating Eq. (2) gives the strain at $\mathbf{x}$, and the pair $\psi = (u, \varepsilon)$, nine components with the strain in engineering Voigt form, satisfies one Lippmann–Schwinger equation with a 9×9 kernel and the contrast operator $\mathbf{\Delta} = \text{blockdiag}(\omega^2\Delta\rho \mathbf{I}_3, \Delta\mathcal{C})$, $\Delta\mathcal{C}$ the Voigt form of δc [8].

The strain model. Table 1 sets out the three representations of the strain that this series meets. They differ in the unknowns, and the test of the equation follows from the unknowns: derivatives at a point are fixed by imposing the equation and its derivatives there, moments over a cell by testing against the cell's own polynomials. This paper is about the third.

Table 1. How the strain is represented within a cell, and the test that follows. The words *collocation* and *Galerkin* describe only the test.

strain model	unknowns of a cell	strain and displacement	test
uniform (the original approximation)	displacement and strain at the centre	strain uniform	at the centre
derived	$\partial^p u$ at the centre, $ P \leq \ell$	strain the gradient of a Taylor displacement, degree $\ell - 1$	the equation and its derivatives at the centre
independent (this paper)	Legendre moments of displacement and strain to degree p	independent, both degree p	against the cell's polynomials (Galerkin)

Table 2. Derived against independent strain, on a layer of thickness D with an exact solution (P wave at normal incidence, $k_p D = 0.6$, $\Delta(\lambda + 2\mu) = 4 \text{ GPa}$, $\Delta\rho = 100 \text{ kg m}^{-3}$): relative error of the reflected displacement with 16 and 32 cells, and the order between them. All three schemes use Legendre polynomials of degree p in each cell, tested against the same. *Independent*: displacement and strain each of degree p . *Derived*: displacement of degree p , strain its derivative in each cell. *Derived, with faces*: the same, with the jumps of the displacement across the faces between cells added to the stiffness source as plane deltas, which makes the derivative that of a distribution.

p	independent			derived			derived, with faces		
	$n = 16$	32	order	16	32	order	16	32	order
1	2.6×10^{-9}	1.6×10^{-10}	4.00	9.0×10^{-5}	2.2×10^{-5}	2.00	1.4×10^{-4}	3.6×10^{-5}	1.98
2	2.6×10^{-14}	round-off	6.00 ^a	1.4×10^{-5}	3.5×10^{-6}	2.00	7.7×10^{-7}	9.5×10^{-8}	3.01
3	round-off	round-off	7.96 ^b	8.2×10^{-10}	5.1×10^{-11}	4.00	1.4×10^{-9}	8.8×10^{-11}	3.98

^a from 4 to 8 cells. ^b from 2 to 4 cells; finer grids are at round-off.

The derived strain loses most of the order: at 16 cells its errors are 3×10^4 times those of

the independent strain for $p = 1$ and 5×10^8 times for $p = 2$ (`scripts/measure_derived_vs_independent_strain.py`). Restoring the jumps of the cellwise displacement across the faces does not recover it, so the loss is not the faces' alone; with the strain an unknown of its own, tested against the same polynomials as the displacement, the scheme reaches the order $2p + 2$ that a Galerkin scheme gives a functional of its solution [8].

The linear Legendre cell. A cube $[-h, h]^3$ with local coordinates $\xi = (\mathbf{x} - \mathbf{x}_c)/h$ carries $\psi(\mathbf{x}) = \sum_b \psi_b L_b(\xi)$ over the four functions $L_b \in \{1, \xi_0, \xi_1, \xi_2\}$, each ψ_b a nine-component vector: 36 unknowns. With the contrast uniform in the cell the source density $\Delta\psi$ is a polynomial on the same four functions. Testing Eq. (2) and its derivative against each L_a gives the single-site equation [9]

$$(\mathbf{M} - \mathbf{K}\Delta)\psi = \langle L, \psi^0 \rangle, \quad \mathbf{M}_{ab} = \langle L_a, L_b \rangle \mathbf{I}_9, \quad (3)$$

with $\mathbf{M} = \text{diag}(8h^3, \frac{8}{3}h^3, \frac{8}{3}h^3, \frac{8}{3}h^3) \otimes \mathbf{I}_9$ and $\mathbf{K}$ the self block, the kernel integrated against L_a over the cell as receiver and L_c over the cell as source. The single-site matrix maps the Legendre coefficients of the incident field to the source moments $\sigma_a = \langle L_a, \Delta\psi \rangle$, the mean force and stress and their first moments,

$$\mathbf{T}_{36} = \mathbf{M}\Delta(\mathbf{M} - \mathbf{K}\Delta)^{-1}\mathbf{M} = \mathbf{M}\Delta(\mathbf{I} - \mathbf{N}\Delta)^{-1}, \quad \mathbf{N} = \mathbf{M}^{-1}\mathbf{K}. \quad (4)$$

Everything specific to the cube is in $\mathbf{N}$; the contrast enters only through Δ .

3 The self block as a series in the wavenumber

The series. The Green's tensor is $\mathbf{G} = (4\pi\mu)^{-1}[\delta_{ij}g_s + k_s^{-2}\partial_i\partial_j(g_s - g_p)]$ with $g_k = e^{ikr}/r = \sum_t (ik)^t r^{t-1}/t!$, so it is a sum over the powers r^m , $m \geq -1$, of two families, $\delta_{ij}r^m$ and $\partial_i\partial_j r^m$ [9]. The propagator adds up to two derivatives. Each entry of the self block is therefore a combination, with 9×9 tables $\mathbf{TA}_t$ and $\mathbf{TB}_t$ fixed by the propagator's structure, of the universal moments of the self cell

$$U_{ac}(m, \iota) = \int W_{ac}(\mathbf{s}) \partial^t r^m \big|_{\mathbf{s}} d^3s, \quad h = 1, \quad (5)$$

W_{ac} the cross-correlation of L_a and L_c over the cell, $\mathbf{s}$ the separation of a receiving and a radiating point, and ι a multi-index of at most four derivatives. Collecting the powers of the wavenumber,

$$\mathbf{K} = \sum_{n \geq 0} (k_s h)^n \mathbf{K}_n, \quad 4\pi\mu \mathbf{K}_n = i^n \left[\frac{\mathbf{A}_n}{n!} - (1 - \gamma^{n+2}) \frac{\mathbf{B}_n}{(n+2)!} \right], \quad \gamma = \frac{k_p}{k_s} = \frac{\beta}{\alpha}, \quad (6)$$

$$\mathbf{A}_n = \sum_{\iota} \mathbf{TA}_{\iota} h^{5-|\iota|} U(n-1, \iota), \quad \mathbf{B}_n = \sum_{\iota} \mathbf{TB}_{\iota} h^{7-|\iota|} U(n+1, \iota). \quad (7)$$

The term $n = 0$ is the static (Kelvin) block. The series is that of e^{ikr} and converges for every k ; at $k_s h \leq 0.2$ seven terms leave less than 4×10^{-9} of the T -matrix (§6). An entry of row type r and column type c (0 for displacement or force, 1 for strain or moment) scales as h^{5-r-c} , so the coefficients need be computed once, at $h = 1$.

The moments, exactly. For odd m the kernel r^m is singular or not smooth at $\mathbf{s} = 0$, which for the self cell is the centre of the support. The integral of Eq. (5) is then reduced, exactly as for two cells that touch [9]: the derivatives that make the kernel a distribution are moved onto W , whose second derivative carries plane deltas at its kinks; W is a polynomial on each of the eight boxes $[0, \pm 2]^3$ that meet at the singular point; and each piece is a box or face integral of a monomial times a power of the distance, reduced from volume to faces to edges by recurrences whose base cases are elementary. Carried out in exact arithmetic, every such moment is a closed form. For even m the kernel is a polynomial, $\partial^t r^m = 0$ for $|t| > m$, and the box integrals are rational. Through $(k_s h)^6$ the self block needs 223 moments, $m = -1$ to 7; each is derived symbolically (`scripts/derive_legendre_cell_dynamic.py`).

4 Six constants, and a rational radiation

The moments of Eq. (5) for odd m are built from the master integrals of the self cell's boxes, whose sides are 2 and whose corners lie at 0 and ± 2 . Their recurrences only multiply powers of $\sqrt{a^2 + b^2 + c^2}$ with $a, b, c \in \{0, 2\}$, which gives $\sqrt{2}$ and $\sqrt{3}$, and end in three base cases: $\operatorname{arsinh}(c/\sqrt{a^2 + b^2})$, which gives $\operatorname{arsinh} 1 = \ln(1 + \sqrt{2})$ and

$$\operatorname{arsinh} \frac{1}{\sqrt{2}} = \ln \frac{1 + \sqrt{3}}{\sqrt{2}} = \frac{1}{2} \ln(2 + \sqrt{3}), \quad (8)$$

since $(1 + \sqrt{3})^2/2 = 2 + \sqrt{3}$, and $\arctan(bc/a\sqrt{a^2 + b^2 + c^2}) = \pi/6$. So for every odd m , and for every derivative, $U(m, \iota)$ is a combination with rational coefficients of

$$1, \quad \pi, \quad \sqrt{2}, \quad \sqrt{3}, \quad \ln(1 + \sqrt{2}), \quad \ln(2 + \sqrt{3}). \quad (9)$$

The two logarithms are those of the fundamental units $1 + \sqrt{2}$ and $2 + \sqrt{3}$ of the real quadratic fields $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{3})$, the fields of the face and body diagonals of the cube; the second is the $-\ln(2 - \sqrt{3})$ of the moments of the expansion about the centre [8]. We write it as $\ln(2 + \sqrt{3})$ because that form is the better conditioned in floating point: $2 - \sqrt{3}$ is formed by a cancellation that magnifies the rounding of $\sqrt{3}$ by $\sqrt{3}/(2 - \sqrt{3}) \approx 6.5$, while $2 + \sqrt{3}$ involves none. In double precision $\frac{1}{2} \ln(2 + \sqrt{3})$ is correct to 1.0×10^{-16} and $-\frac{1}{2} \ln(2 - \sqrt{3})$ to 2.7×10^{-16} . By Eq. (7), $\mathbf{K}_n$ draws on $m = n - 1$ and $n + 1$. An even power of $k_s h$ therefore uses odd m and is a combination of the six constants; an odd power uses even m , polynomial kernels, and is rational. Divided by the 4π of Eq. (6), an even power is a rational combination of $1/\pi$, 1 , $\sqrt{2}/\pi$, $\sqrt{3}/\pi$, $\ln(1 + \sqrt{2})/\pi$ and $\ln(2 + \sqrt{3})/\pi$, and an odd power a rational multiple of $1/\pi$. The derivation confirms both through the highest power computed: the constants that occur in $\mathbf{K}_0$, $\mathbf{K}_2$, $\mathbf{K}_4$ and $\mathbf{K}_6$ are exactly the six, and every entry of $\mathbf{K}_1$, $\mathbf{K}_3$ and $\mathbf{K}_5$ is rational. In the static strain blocks the two logarithms occur only in the combination $\Lambda = [\ln(1 + \sqrt{2}) - \frac{1}{2} \ln(2 + \sqrt{3})]/\pi$ of Nestor [9]; in the displacement blocks and at the dynamic orders they occur separately.

The radiation. The odd powers are the radiation of the cell. The first, $n = 1$, has a single non-zero entry: the uniform displacement against the mean force, the cell's monopole. In the

reduced form of §5 it is

$$N_{T_{1u},1}[u, u] = \frac{i}{\mu} \left[\frac{2}{\pi} - (1 - \gamma^3) \frac{4}{3! \pi} \right] = \frac{2i(2 + \gamma^3)}{3\pi\mu}, \quad (10)$$

so that the term $k_s h^2 N_{T_{1u},1}$ of $\mathbf{M}^{-1} \mathbf{K}$ is $8h^3$ times $i k_s (2 + \gamma^3)/(12\pi\mu)$, the imaginary part of $\mathbf{G}(0)$: the radiation damping of a point force carrying the cell's total force, as it must be at lowest order. The next odd power, $n = 3$, is again rational; it enters A_{1g} , E_g , T_{1g} , T_{2g} and T_{1u} , and vanishes on T_{2u} , A_{2u} and E_u ; the fifth enters every representation.

5 The cube group

With the contrast uniform in the cell, the 48 symmetries of the cube (the signed permutations of its axes, the group O_h) act on the 36 unknowns, and the self block, the Gram matrix and the contrast all commute with that action: the cube is mapped onto itself, the isotropic Green's tensor is covariant, and the uniform contrast is invariant. By Schur's lemma the single-site system then splits exactly into independent blocks, one for each irreducible representation Γ of O_h that occurs,

$$\mathbf{K} = \bigoplus_{\Gamma} (\mathbf{K}_{\Gamma} \otimes \mathbf{I}_{d_{\Gamma}}), \quad (11)$$

an $n_{\Gamma} \times n_{\Gamma}$ block, n_{Γ} the number of times Γ occurs, repeated for each of the d_{Γ} rows of Γ ; likewise for $\mathbf{M}$, $\mathbf{\Lambda}$ and $\mathbf{T}_{36}$. The commutation holds for the complex self block at every frequency, radiation included, so the splitting holds power by power. The displacement's first moments $\xi_i u_j$ (9 unknowns) span $A_{1g} + E_g + T_{1g} + T_{2g}$; the uniform strain (6) spans $A_{1g} + E_g + T_{2g}$; the uniform displacement (3) spans T_{1u} ; and the strain's first moments (18) span $3T_{1u} + 2T_{2u} + A_{2u} + E_u$. Altogether there are 15 even unknowns, $2A_{1g} + 2E_g + T_{1g} + 2T_{2g}$, and 21 odd, $4T_{1u} + 2T_{2u} + A_{2u} + E_u$; even and odd never couple, and the largest block is T_{1u} , 4×4 . Written out, with V the matrix whose columns are the partner vectors (Appendix A),

$$\begin{aligned} V^{-1} \mathbf{T}_{36} V = & \underbrace{\mathbf{T}_{A_{1g}}}_{2 \times 2} \oplus \underbrace{(\mathbf{T}_{E_g} \otimes \mathbf{I}_2)}_{2 \times 2} \oplus \underbrace{(\mathbf{T}_{T_{1g}} \otimes \mathbf{I}_3)}_{1 \times 1} \oplus \underbrace{(\mathbf{T}_{T_{2g}} \otimes \mathbf{I}_3)}_{2 \times 2} \quad (\text{even, 15}) \\ & \oplus \underbrace{(\mathbf{T}_{T_{1u}} \otimes \mathbf{I}_3)}_{4 \times 4} \oplus \underbrace{(\mathbf{T}_{T_{2u}} \otimes \mathbf{I}_3)}_{2 \times 2} \oplus \underbrace{\mathbf{T}_{A_{2u}}}_{1 \times 1} \oplus \underbrace{(\mathbf{T}_{E_u} \otimes \mathbf{I}_2)}_{1 \times 1} \quad (\text{odd, 21}), \end{aligned} \quad (12)$$

each block given by Eq. (13) below and $\mathbf{I}_d$ repeating it once for each row of its representation. The reduction does not make the matrix hold fewer numbers: in the natural ordering the 180 non-zero entries of $\mathbf{M} - \mathbf{K}\mathbf{\Lambda}$ take only 25 distinct values up to sign, the symmetry of the cube repeating them within and between the blocks of the four functions, while the eight blocks hold 35 entries, combinations of those values in the partner basis. What it changes is the coupling: the 36 unknowns, coupled through those entries in one 36×36 system, separate into independent systems of at most four unknowns (Figure 1), each inverted on its own, which is what lets $\mathbf{T}_{36}$ be written out in closed form.

Appendix A gives the group's action on the unknowns, why the commutation holds, Schur's lemma and what it requires, and the vectors that span one row of each block (Table 4); the

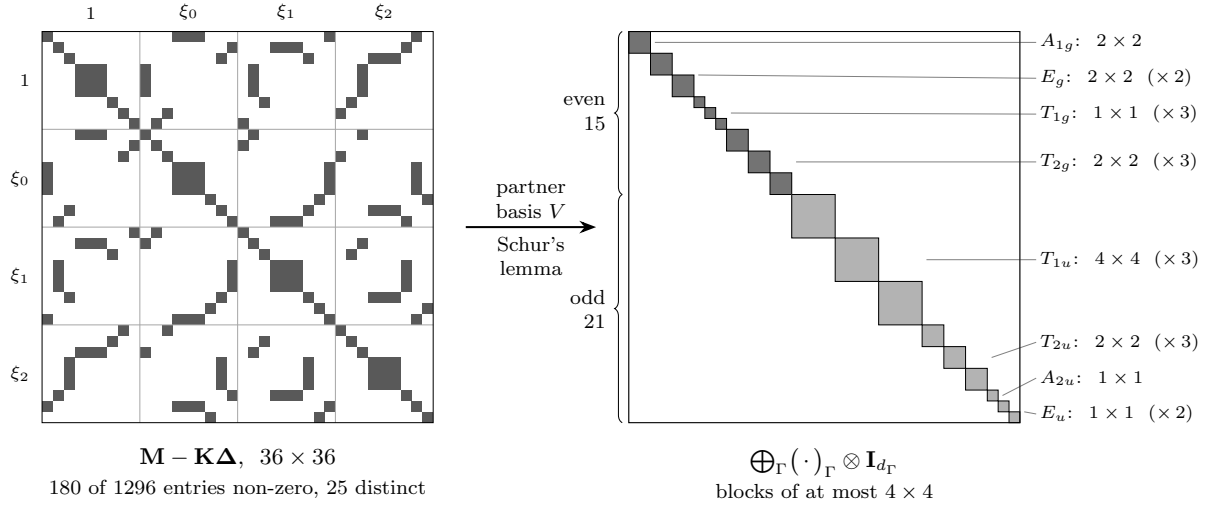


Figure 1. The single-site system before and after the change to the partner basis. Left: the non-zero entries of $\mathbf{M} - \mathbf{K}\Delta$ (dark) in the natural ordering, the four functions $1, \xi_0, \xi_1, \xi_2$ each carrying the nine components, at $k_S h = 0.1$; 180 of the 1296 entries are non-zero, and they take only 25 distinct values up to sign, the symmetry of the cube repeating them within and between the blocks of the functions. Right: the same matrix in the basis of partner vectors (Table 4), by Schur's lemma block diagonal, Eq. (12): each irreducible representation gives one block, of size its multiplicity, repeated once for each of its rows; the 15 even unknowns (darker) never couple to the 21 odd (lighter). The same structure holds at every frequency and for every power of $k_S h$, and for $\mathbf{T}_{36}$.

reduction is exact, with no residual at any power computed (§6).

On each representation

$$\mathbf{T}_{\Gamma} = M_{\Gamma} \Delta_{\Gamma} (\mathbf{I} - \mathbf{N}_{\Gamma} \Delta_{\Gamma})^{-1}, \quad \mathbf{N}_{\Gamma} = \frac{1}{\mu} \sum_{n \geq 0} (k_S h)^n \mathbf{i}^n \left[\frac{\mathbf{P}_{\Gamma, n}}{n!} - (1 - \gamma^{n+2}) \frac{\mathbf{Q}_{\Gamma, n}}{(n+2)!} \right], \quad (13)$$

with Δ_{Γ} the contrast on the vectors of Γ , which is $\omega^2 \Delta \rho$ on a displacement vector and a combination of $\Delta \lambda$ and $\Delta \mu$ on a strain vector, and $\mathbf{P}_{\Gamma, n}, \mathbf{Q}_{\Gamma, n}$ the reduced forms of $\mathbf{A}_n$ and $\mathbf{B}_n$ divided by $4\pi M_{\Gamma}$. Each of their entries is a rational vector on the six constants (§4); all of them, through the highest power computed, are in `data/self_blocks_irreps.txt`. As an example, the uniform dilatation (A_{1g} , the strain vector) has $P_0 = -\frac{1}{3}$ and $Q_0 = -\frac{2}{3}$, so that its static entry is $\mu N = -\frac{1}{3} + \frac{1}{2}(1 - \gamma^2)\frac{2}{3} = -\gamma^2/3$, that is $N = -1/[3(\lambda + 2\mu)]$, the value for every shape [9]; all 36 static entries of the uniform and linear strain reproduce those of Nestor [9], coefficient by coefficient on its five constants. Its first dynamic correction is

$$\mu N_{A_{1g}, 2}[\varepsilon, \varepsilon] = -\frac{1}{2} \left[P_2 - \frac{1 - \gamma^4}{12} Q_2 \right], \quad \pi P_2 = \frac{4}{15} - \frac{4}{9} \pi + \frac{4}{15} \sqrt{2} - \frac{8}{15} \sqrt{3} + \frac{4}{3} \ln(1 + \sqrt{2}) + \frac{4}{3} \ln(2 + \sqrt{3}), \quad (14)$$

and likewise for Q_2 (the data file); $(k_S h)^2$ times this is the leading change with frequency of the cell's dilatational self-interaction.

Inversion in closed form. The reduction matters because of what it does to the inverse in Eq. (4). A 36×36 inverse is computed numerically, one contrast and one frequency at a time; an

inverse of order at most four is written down. On each representation the contrast is the same simple matrix whatever the cell's material, from Table 4 and the contrast operator of §2:

$$\begin{aligned}\Delta_{T_{1g}} &= \omega^2 \Delta \rho, & \Delta_{A_{2u}} &= \Delta_{E_u} = 2\Delta\mu, & \Delta_{T_{2u}} &= 2\Delta\mu \mathbf{I}_2, \\ \Delta_{A_{1g}} &= \text{diag}(\omega^2 \Delta \rho, 3\Delta\lambda + 2\Delta\mu), & \Delta_{E_g} &= \Delta_{T_{2g}} = \text{diag}(\omega^2 \Delta \rho, 2\Delta\mu), \\ \Delta_{T_{1u}} &= \omega^2 \Delta \rho \oplus \begin{pmatrix} \Delta\lambda + 2\Delta\mu & 2\Delta\lambda \\ \Delta\lambda & 2\Delta\lambda + 2\Delta\mu \end{pmatrix} \oplus 2\Delta\mu.\end{aligned}\tag{15}$$

With $\mathbf{X}_\Gamma = \mathbf{N}_\Gamma \Delta_\Gamma$, Eq. (13) is inverted by Cramer's rule,

$$\mathbf{T}_\Gamma = \frac{M_\Gamma \Delta_\Gamma \text{adj}(\mathbf{I} - \mathbf{X}_\Gamma)}{\det(\mathbf{I} - \mathbf{X}_\Gamma)}, \quad \text{for instance} \quad (\mathbf{I} - \mathbf{X})^{-1} = \frac{1}{(1 - X_{11})(1 - X_{22}) - X_{12}X_{21}} \begin{pmatrix} 1 - X_{22} & X_{12} \\ X_{21} & 1 - X_{11} \end{pmatrix}\tag{16}$$

for a block of order two, and $M\Delta/(1 - N\Delta)$ for a block of order one; the block of order four, T_{1u} , has a determinant of degree four in the contrast. Three things follow, none of which a numerical inverse gives.

The T-matrix is an explicit function of the contrast. Each entry of $\mathbf{T}_\Gamma$ is a ratio of polynomials in $\Delta\lambda$, $\Delta\mu$ and $\omega^2 \Delta \rho$, of degree at most n_Γ , with coefficients the entries of $\mathbf{N}_\Gamma$, which carry the frequency through Eq. (13) and nothing else of the material. A new contrast needs no new solve, and the derivatives of $\mathbf{T}$ with respect to the contrast, which an inversion for the medium needs, are those of explicit rational functions.

Its poles are written down. $\mathbf{T}_\Gamma$ is singular where $\det(\mathbf{I} - \mathbf{N}_\Gamma \Delta_\Gamma) = 0$, one condition per representation, explicit in the contrast; and the expansion of $(\mathbf{I} - \mathbf{X}_\Gamma)^{-1}$ in powers of the contrast, the Born series of the cell, converges exactly when the eigenvalues of the at most 4×4 matrix $\mathbf{X}_\Gamma$ lie inside the unit circle.

Classical results appear as special cases. In the static limit the uniform dilatation decouples ($\omega^2 \Delta \rho \rightarrow 0$ removes the displacement row of A_{1g}), $N = -1/[3(\lambda + 2\mu)]$, and with $D = 3\Delta\lambda + 2\Delta\mu = 3\Delta K$, $K = \lambda + \frac{2}{3}\mu$ the bulk modulus and $K' = K + \Delta K$,

$$T_{A_{1g}} = \frac{8h^3 D}{1 + D/[3(\lambda + 2\mu)]} = 8h^3 3\Delta K \frac{K + \frac{4}{3}\mu}{K' + \frac{4}{3}\mu},\tag{17}$$

Eshelby's dilatational result for a sphere [5], with its pole at $K' = -\frac{4}{3}\mu$; the closed form agrees with the numerical $\mathbf{T}_{36}$ to 1.1×10^{-15} (§6). The two shear channels E_g and T_{2g} take the same form with the cube's constants of Nestor [9] in place of the sphere's.

6 Checks

Every step is checked against an evaluation that shares none of it (`scripts/derive_legendre_cell_dynamic.py`), with the background $\alpha = 5000 \text{ m s}^{-1}$, $\beta = 3000 \text{ m s}^{-1}$, $\rho = 2500 \text{ kg m}^{-3}$ and the contrast $\Delta\lambda = 2 \text{ GPa}$, $\Delta\mu = 1 \text{ GPa}$, $\Delta\rho = 100 \text{ kg m}^{-3}$. Table 3 gives the results.

The differences fall as $(k_s h)^7$, the first power omitted (and as $(k_s h)^5$ when the series stops at

Table 3. Checks of the closed form through $(k_S h)^6$. The reference for the series is the package’s series of the self block (master integrals at forty digits for odd m , Gauss rules for even m), which itself agrees with quadrature of the singular integrals to 1.4×10^{-15} .

check	result
223 exact moments, $m = -1$ to 7, against the package’s self block, exact series to $n = 6$, against the full series, $k_S h = 0.02, 0.05, 0.1, 0.2$	1.7×10^{-15} $1.1 \times 10^{-14}, 6.6 \times 10^{-12}, 8.4 \times 10^{-10}, 1.1 \times 10^{-7}$
apparent order of that difference	7.00
$\mathbf{T}_{36}$ from the exact series against $\mathbf{T}_{36}$ by quadrature, $k_S h = 0.05, 0.1, 0.2$	$2.1 \times 10^{-13}, 2.7 \times 10^{-11}, 3.5 \times 10^{-9}$
apparent order	7.00
the same truncated at $n = 4$	$5.9 \times 10^{-10}, 1.9 \times 10^{-8}, 6.1 \times 10^{-7}$; order 5.00
invariance of every representation’s span under every $\mathbf{K}_n$	exact (no residual)
$\mathbf{T}_{36} V_\Gamma = V_\Gamma \mathbf{T}_\Gamma$, Eq. (13), worst over Γ , $k_S h = 0.05, 0.1, 0.2$	$4.4 \times 10^{-13}, 5.6 \times 10^{-11}, 7.2 \times 10^{-9}$
static uniform dilatation	$-1/[3(\lambda + 2\mu)]$ exactly
static dilatational block of $\mathbf{T}_{36}$ against Eshelby’s sphere, Eq. (17)	1.1×10^{-15}
the 36 static strain entries against Nestor [9]	identical coefficients
radiation at $n = 1$	$\text{Im } \mathbf{G}(0)$ exactly, Eq. (10)

$n = 4$), so they are the truncation of the series and not an error of its terms. The closed form is that of the exact self block: what it omits at a given order is stated by the next term, which is itself in closed form.

7 Discussion

A frequency sweep with no integral at any frequency. The single-site T -matrix of the linear Legendre cell is now known in closed form at every power of the frequency, and what that buys is a sweep. The coefficients $\mathbf{K}_n$ depend on neither the frequency, the cell size nor the medium: they are computed once, in exact arithmetic, and stored. At each frequency the self block is then the polynomial $\sum_n (k_S h)^n \mathbf{K}_n$, scaled to the cell by Eq. (6), and the T -matrix one 36×36 solve, or eight solves of at most 4×4 on the irreducible representations. Nothing singular is integrated at any frequency. Measured on one machine (`scripts/measure_legendre_cell_sweep.py`), one frequency costs 0.29 ms by the closed form against 2.0 s by quadrature of the singular self integrals, a factor of about 7000, and against 16 ms by the package’s numerical series of the same moments. The one-off cost of the closed form, the 223 exact moments and their assembly and reduction through $(k_S h)^6$, is 30 s on four cores, paid once for every cell, frequency and medium; against quadrature it is recovered after about fifteen frequencies, and a sweep of a thousand frequencies that quadrature would take half an hour to tabulate costs half a minute in all. The coupling between cells has the same structure [9]: its coefficients too are universal and computed once, and over 1024 frequencies its series costs 6 to 13 min against 34 to 89 min by quadrature on the grids measured there. With both, a model of cells is assembled at any frequency from stored numbers.

The range of the truncated series. The series converges for every $k_S h$, but a sweep evaluates it truncated. The figures that follow are for the single-site T -matrix of one cell, compared entry by entry (largest difference relative to the largest entry) with the same T -matrix built on the self block by quadrature. The cell's interaction with its own field is summed exactly by the inverse in Eq. (4), and no other cell is involved: the truncation is only that of the series in the frequency of the self block. Through $(k_S h)^6$ the T -matrix is accurate to 3.5×10^{-9} at $k_S h = 0.2$, 5.9×10^{-8} at 0.3, 2.1×10^{-6} at 0.5 and 2.6×10^{-4} at 1, against quadrature that agrees with the package's series to 10^{-16} ; stopping at $(k_S h)^4$ instead loses three orders of magnitude at $k_S h = 0.05$ and less than one at $k_S h = 1$. A discretisation with ten cells to the shear wavelength has $k_S h = \pi/10 \approx 0.31$, where the truncation through $(k_S h)^6$ leaves 8.2×10^{-8} . Beyond that, the next powers are in closed form like the first, from the same derivation, at a one-off cost that grows with the number of moments.

What is not yet covered. Two things. The contrast here is uniform in the cell; a graded contrast couples the even and odd unknowns and leaves only the operations that fix its gradient [9], so the reduction is weaker, though the moments are the same and the closed form carries over. And the field is linear; the quadratic cell of 90 unknowns needs the moments of monomials to degree four, which the same derivation provides. The comparison with the single site of the expansion about the centre, whose self-interaction is taken at the centre rather than averaged over the cell [8, 9], is to be condensed here from the earlier paper.

Data availability

The derivation and its checks are `scripts/derive_legendre_cell_dynamic.py`, which writes `data/self_moments_exact.pkl` (every moment, exact) and `data/self_blocks_irreps.txt` (every reduced entry, on the six constants), in the repository <https://github.com/tmnestor/MultipleScatteringCalculations>. The cost of a sweep and the range of the truncated series are measured by `scripts/measure_legendre_cell_sweep.py`. The static blocks are `scripts/derive_graded_voxel_site_blocks.py`.

A The cube group and Schur's lemma

The splitting of §5 rests on Schur's lemma from the representation theory of finite groups [11, 12]. This appendix sets out the group's action on the unknowns of the cell, why the cell's matrices commute with it, and the form the lemma then gives them.

Symmetries of the cube. A *symmetry* of the cube is a linear map $\xi \mapsto Q\xi$ of space that preserves distances and maps the cube $[-1, 1]^3$ onto itself; it fixes the centre, and Q is an orthogonal matrix, $Q^T Q = \mathbf{I}$. Such a map must send the six face centres $\pm \mathbf{e}_j$ to face centres, so each column of Q is some $\pm \mathbf{e}_i$: Q is a *signed permutation matrix*, a permutation of the three axes combined with a choice of sign for each. There are $3! \times 2^3 = 48$ of them, the rotations of the

cube ($\det Q = +1$) and their products with the inversion $\xi \mapsto -\xi$ ($\det Q = -1$), which include the reflections in the coordinate and diagonal planes. The product of two symmetries and the inverse $Q^{-1} = Q^T$ of one are again symmetries, so they form a group, the octahedral group O_h . For example, the quarter turn about axis 0 and the reflection in the plane $\xi_0 = 0$ are

$$Q_{\text{turn}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad Q_{\text{refl}} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (18)$$

How a symmetry acts on the field of the cell. Applying Q to a field carries the value at ξ to $Q\xi$ and turns its components with it: the displacement is a vector, $u \mapsto Qu$, and the strain a symmetric tensor, $\varepsilon \mapsto Q\varepsilon Q^T$, that is $\varepsilon'_{ij} = \sum_{k,l} Q_{ik} Q_{jl} \varepsilon_{kl}$. On the nine components $\psi = (u, \varepsilon)$ this is the 9×9 matrix

$$S(Q) = \begin{pmatrix} Q & 0 \\ 0 & \Sigma(Q) \end{pmatrix}, \quad (19)$$

with $\Sigma(Q)$ the 6×6 matrix of $\varepsilon \mapsto Q\varepsilon Q^T$ in Voigt form. The transformed field is $\psi'(\xi) = S(Q)\psi(Q^T\xi)$. For the cell's field $\psi(\xi) = \sum_b \psi_b L_b(\xi)$ the functions are moved as well: 1 is unchanged, and $L_k(Q^T\xi) = (Q^T\xi)_k = \sum_j Q_{jk} \xi_j$ for the linear functions $L_k = \xi_k$. So the four functions are mixed by the 4×4 matrix

$$T(Q) = \begin{pmatrix} 1 & 0 \\ 0 & Q \end{pmatrix}, \quad (20)$$

and the coefficients of the transformed field are

$$\psi'_a = \sum_{b=0}^3 T_{ab}(Q) S(Q) \psi_b, \quad a = 0, \dots, 3. \quad (21)$$

Two things happen at once and independently: $T(Q)$ mixes *which function* carries the field, and $S(Q)$ mixes *which component* of the field it is.

The Kronecker product. Stack the 36 unknowns as one column, $\Psi = (\psi_0, \psi_1, \psi_2, \psi_3)$, each ψ_b nine components, so that the unknown with function index a and component i sits in row $9a + i$. Eq. (21) is then $\Psi' = D(Q)\Psi$ with the 36×36 matrix

$$D(Q) = T(Q) \otimes S(Q) = \begin{pmatrix} T_{00}S & T_{01}S & T_{02}S & T_{03}S \\ T_{10}S & T_{11}S & T_{12}S & T_{13}S \\ T_{20}S & T_{21}S & T_{22}S & T_{23}S \\ T_{30}S & T_{31}S & T_{32}S & T_{33}S \end{pmatrix}, \quad D(Q)_{9a+i, 9b+j} = T_{ab}(Q) S_{ij}(Q), \quad (22)$$

the Kronecker product: the 4×4 array of 9×9 blocks whose block (a, b) is $S(Q)$ scaled by the number $T_{ab}(Q)$. It is the matrix of a map that acts on the function index by T and on the component index by S , each without regard to the other. For the reflection of Eq. (18), $T = \text{diag}(1, -1, 1, 1)$, and S is diagonal with -1 on u_0 and on the shears ε_{01} and ε_{02} , which change sign with ξ_0 , and $+1$ elsewhere; so D is diagonal, and the sign of the unknown $\xi_0 u_0$, for instance,

is $T_{11}S_{00} = (-1)(-1) = +1$. Because the Kronecker product satisfies $(A \otimes B)(C \otimes F) = AC \otimes BF$, and T and S each respect products,

$$D(Q_1)D(Q_2) = D(Q_1Q_2) \quad \text{for all } Q_1, Q_2 \in O_h : \quad (23)$$

the assignment $Q \mapsto D(Q)$ turns composition of symmetries into multiplication of matrices, which is what is meant by a *representation* of the group. For a signed permutation, $Q\epsilon Q^T$ permutes the strain components with signs and sends shears to shears, so $\Sigma(Q)$, $S(Q)$, $T(Q)$ and $D(Q)$ are all signed permutation matrices, and orthogonal: $D(Q)^{-1} = D(Q)^T = D(Q^T)$.

Why the cell's matrices commute with it. Three facts make

$$D(Q)\mathbf{K}D(Q)^T = \mathbf{K}, \quad D(Q)\mathbf{M}D(Q)^T = \mathbf{M}, \quad D(Q)\mathbf{\Delta}D(Q)^T = \mathbf{\Delta} \quad \text{for all 48 } Q. \quad (24)$$

The domain is invariant: Q maps the cube onto itself. *The kernel is covariant:* the background is isotropic, so its Green's tensor satisfies $\mathbf{G}(Q\mathbf{r}) = Q\mathbf{G}(\mathbf{r})Q^T$ for every orthogonal Q , reflections included, and its derivatives transform as tensors of the corresponding rank; changing variables $\mathbf{x} \mapsto Q\mathbf{x}$, $\mathbf{y} \mapsto Q\mathbf{y}$ in the double integral of $\mathbf{K}$ then gives the first identity. *The contrast is invariant:* uniform in the cell, $\omega^2\Delta\rho$ is a scalar and δc an isotropic tensor, so $\mathbf{\Delta}$ commutes with $S(Q)$; and the Gram matrix $\mathbf{M}$ only weights the functions 1 and ξ_i , which Q permutes among themselves with equal weights. The system matrix $\mathbf{M} - \mathbf{K}\mathbf{\Delta}$, its inverse, and $\mathbf{T}_{36}$ of Eq. (4) therefore commute with every $D(Q)$. Nothing here requires $\mathbf{K}$ to be Hermitian or real: the identities hold for the complex self block at every frequency, radiation included, and power by power for each $\mathbf{K}_n$.

Schur's lemma. A representation is *irreducible* when no subspace other than 0 and the whole is mapped into itself by every $D(Q)$. Schur's lemma states that a linear map between two irreducible representations that commutes with the group action is zero if the representations are inequivalent, and a multiple of the identity if they are the same. The 36-dimensional representation of Eq. (22) is not irreducible; it is a direct sum of copies of the ten irreducible representations of O_h , labelled by their dimension (A : 1, E : 2, T : 3) and by their parity under the inversion $\xi \mapsto -\xi$ (g even, u odd). How many copies n_Γ of each Γ occur follows from the characters, the traces of the representation matrices: $n_\Gamma = \frac{1}{48} \sum_Q \chi_\Gamma(Q) \text{tr} D(Q)$, with $\text{tr} D(Q) = \text{tr} T(Q) \text{tr} S(Q)$. Every irreducible representation of O_h can be written with real matrices and has real characters, so the lemma holds over the real numbers, and applies to the real and imaginary parts of $\mathbf{K}$ alike. Applied to an operator that commutes with $D(Q)$, it has two consequences. First, the operator does not couple unknowns of different Γ . Second, within the n_Γ copies of a Γ of dimension d_Γ , if a basis is chosen in which every copy is represented by the *same* matrices (each basis vector a *partner*, transforming as a given row of Γ), the operator acts as an $n_\Gamma \times n_\Gamma$ matrix on the multiplicity index and as the identity on the row index, which is Eq. (11), and likewise for $\mathbf{M}$, $\mathbf{\Delta}$, $\mathbf{M} - \mathbf{K}\mathbf{\Delta}$ and $\mathbf{T}_{36}$. The block $\mathbf{K}_\Gamma$ is then found from any one row: with V_Γ the $36 \times n_\Gamma$ matrix whose columns are the partners of that row, $\mathbf{K}V_\Gamma = V_\Gamma\mathbf{K}_\Gamma$. Inverting the 36×36 system reduces to inverting $n_\Gamma \times n_\Gamma$ blocks, of size at most 4×4 .

The partners. Table 4 gives the partners of one row of each Γ as integer vectors. They are chosen so that within each Γ they transform alike: the displacement's first moment $\xi_1 u_2 + \xi_2 u_1$ and the shear $2\varepsilon_{12}$, for instance, both transform as the $(1, 2)$ row of T_{2g} . The reduction is then exact: $\mathbf{K}_n V_\Gamma = V_\Gamma \mathbf{K}_{\Gamma,n}$ with no residual at any power computed (§6).

Table 4. One row of each irreducible representation, as combinations of the unknowns $\xi_i u_j$ ($L_b = \xi_i$ times the displacement component j), u_j , ε_{ij} and $\xi_i \varepsilon_{jk}$, axes $0, 1, 2$.

Γ	vectors	M_Γ/h^3
A_{1g}	$\xi_0 u_0 + \xi_1 u_1 + \xi_2 u_2; \quad \varepsilon_{00} + \varepsilon_{11} + \varepsilon_{22}$	$\frac{8}{3}, 8$
E_g	$2\xi_0 u_0 - \xi_1 u_1 - \xi_2 u_2; \quad 2\varepsilon_{00} - \varepsilon_{11} - \varepsilon_{22}$	$\frac{8}{3}, 8$
T_{1g}	$\xi_1 u_2 - \xi_2 u_1$	$\frac{8}{3}$
T_{2g}	$\xi_1 u_2 + \xi_2 u_1; \quad 2\varepsilon_{12}$	$\frac{8}{3}, 8$
T_{1u}	$u_0; \quad \xi_0 \varepsilon_{00}; \quad \xi_0(\varepsilon_{11} + \varepsilon_{22}); \quad 2(\xi_1 \varepsilon_{01} + \xi_2 \varepsilon_{02})$	$8, \frac{8}{3}, \frac{8}{3}, \frac{8}{3}$
T_{2u}	$\xi_0(\varepsilon_{11} - \varepsilon_{22}); \quad 2(\xi_1 \varepsilon_{01} - \xi_2 \varepsilon_{02})$	$\frac{8}{3}, \frac{8}{3}$
A_{2u}	$2(\xi_0 \varepsilon_{12} + \xi_1 \varepsilon_{02} + \xi_2 \varepsilon_{01})$	$\frac{8}{3}$
E_u	$2(\xi_1 \varepsilon_{02} - \xi_2 \varepsilon_{01})$	$\frac{8}{3}$

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